

Deep Learning for NLP

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1. Abstract

This report documents the Homework 4 system for response clarity classification on QEvasion [1]. The target labels remain Clear Reply, Ambivalent, and Clear Non-Reply, but the method changes from the single-invocation prompting of Homework 3 to the required D3-Agentic Prompting architecture. D3 decomposes classification into four specialized Qwen agents: Question Intent, Answer Content, Gap and Evasion, and Decision.

The implementation uses Qwen/Qwen3.5-0.8B [4] for the required core system and compares it with a same-model single-agent prompt. This keeps the experimental contrast focused on decomposition rather than model scale. The notebook and script select the final submitted system by validation macro F1, using accuracy only as a tie-breaker.

The final local validation run is a negative result. The same-model single-agent comparator reaches 0.3667 accuracy and 0.3062 macro F1 on a balanced 30-example validation sample, while the required D3-agentic system reaches 0.2667 accuracy and 0.2213 macro F1. D3 exposes useful intermediate traces and recovers some Ambivalent cases, but the decomposition does not improve the final classifier under the fixed small-model setting.

2. Task and Prior Assignments

All four assignments solve the same three-way response clarity task. Homework 1 used classical vector-based classifiers, Homework 2 used encoder-only transformer fine-tuning, Homework 3 used single-invocation Qwen prompting, and Homework 4 uses a lightweight multi-agent Qwen pipeline.

Homework	System	Evaluation	Accuracy	Macro F1	Note
HW1	TF-IDF + Logistic Regression	official test	~0.63	~0.45	Best classical baseline
HW1	GloVe Wiki + Logistic Regression	official test	–	~0.39	Mean-pooled embedding baseline
HW1	GloVe Twitter + Logistic Regression	official test	–	~0.38	Conversational embedding baseline
HW2	BERT-base fine-tuning	official test	0.6299	0.5561	Best encoder-only transformer
HW2	DistilBERT fine-tuning	official test	0.5779	0.5066	Smaller encoder-only transformer
HW2	DeBERTa-v3 fine-tuning	official test	0.6688	0.2672	Collapsed to all Ambivalent predictions
HW3	Qwen 4B few-shot prompting	balanced validation	0.4667	0.4667	Best single-invocation prompting run
HW3	Qwen 0.8B few-shot prompting	balanced validation	0.4667	0.4091	Same model scale as HW4 core D3
HW4	Qwen 0.8B single-agent	balanced validation	0.3667	0.3062	Selected by macro F1 for submission
HW4	Qwen 0.8B D3-agentic	balanced validation	0.2667	0.2213	Required four-agent D3 system

Table 1: Cross-assignment comparison. HW1 and HW2 numbers are official-test results from the earlier reports; HW3 and HW4 numbers are balanced-validation diagnostic runs and are not directly official-test comparable.

The central recurring difficulty is Ambivalent. Classical methods confuse it with both endpoint classes, single-prompt Qwen systems often fail to use it, and unstable fine-tuning can collapse into predicting only Ambivalent. This motivates macro F1 and concrete error analysis.

3. D3-Agentic Prompting Design

The D3 pipeline is not repeated voting. Each agent performs a distinct linguistic sub-task and passes its output to later stages.

1. **Question Intent Agent:** identifies what the journalist asks for and what information a clear answer would need.
2. **Answer Content Agent:** extracts what the answer actually says, including explicit claims, relevant details, hedges, and omissions.
3. **Gap and Evasion Agent:** compares expected information with actual answer content and identifies missing requirements or evasion patterns.
4. **Decision Agent:** maps the previous agents' outputs to exactly one final clarity label.

All core agents use Qwen/Qwen3.5-0.8B. This is the required model in the specification and avoids a confound where one agent performs better merely because it uses a larger model. The single-agent comparator also uses the same model and taxonomy so that the comparison measures the effect of decomposition.

4. Prompt Format

The prompts include the original question and answer, the dataset label taxonomy, and compact JSON output constraints. The taxonomy is:

Clear Reply $\leftarrow$ Explicit,
 Ambivalent $\leftarrow$ Implicit, Dodging, General, Deflection, Partial/half-answer,
 Clear Non-Reply $\leftarrow$ Declining to answer, Claims ignorance, Clarification.

The first three agents are instructed not to emit a final label. Only the Decision Agent may output Clear Reply, Ambivalent, or Clear Non-Reply. This design supports analysis: the generated `agent_outputs.csv` files expose whether an error originates from misunderstanding the question, missing answer content, weak gap analysis, or the final mapping step.

5. Experimental Protocol

The implementation uses deterministic decoding, a fixed random seed, and a balanced validation sample of 10 examples per class by default. The main command is:

```
python -m artificial_intelligence.homeworks.hw20260609.sdi2200160 \
  --preset full --eval-per-label 10
```

The Kaggle notebook is named:

`sdi2200160 homework-4-d3-agentic.ipynb`

The private Kaggle notebook link will be inserted after upload.

It writes:

`submission_best_d3_agentic_system.csv`

The full validation comparison should include:

- d3-agentic: required four-agent system;
- single-agent: same-model direct prompt comparator;
- previous baselines from Table 1.

6. Validation Results

Table 2 reports the completed full run. The validation set is balanced by construction: 10 examples from each of Clear Reply, Ambivalent, and Clear Non-Reply. The selected system is qwen-0.8b_single-agent, because it has the higher macro F1. The required D3 system is retained in the analysis because it is the assigned method and because its intermediate traces are useful for diagnosing failure modes.

System	Calls	Accuracy	Macro F1	Precision	Recall	Invalid Rate
Single-agent Qwen 0.8B	1	0.3667	0.3062	0.4400	0.3667	0.0000
D3-agentic Qwen 0.8B	4	0.2667	0.2213	0.2164	0.2667	0.1667

Table 2: HW4 validation results from runs_hw4_full/experiment_summary.csv.

True \ Pred.	Clear Reply	Ambivalent	Clear Non-Reply
Clear Reply	8	2	0
Ambivalent	10	0	0
Clear Non-Reply	7	0	3

Single-agent

True \ Pred.	Clear Reply	Ambivalent	Clear Non-Reply	Invalid
Clear Reply	2	6	0	2
Ambivalent	3	6	0	1
Clear Non-Reply	1	7	0	2

D3-agentic

Table 3: Validation confusion matrices. D3 recovers more Ambivalent examples than the direct prompt, but it never correctly predicts Clear Non-Reply in this run and emits invalid labels on 5 of 30 cases.

7. Error Analysis

The direct prompt has the better aggregate score, but its behavior is still very poor. It predicts Clear Reply 25 times, Ambivalent only twice, and Clear Non-Reply three times. Its Ambivalent recall is exactly 0.0: every true Ambivalent example is mapped to Clear Reply. This is the same middle-class failure pattern observed in the earlier Qwen prompting assignment, where the model had trouble preserving the distinction between an incomplete answer and a clearly answered question.

The D3 system fails differently. It predicts Ambivalent 19 times, Clear Reply six times, and produces five invalid labels. It correctly recovers six of the ten Ambivalent examples, so the decomposition does make the model more willing to use the middle class. However, that gain is overwhelmed by two costs: all ten Clear Non-Reply examples are missed, and invalid Decision Agent outputs reduce the valid rate to 0.8333. The Gap and Evasion stage often recognizes that an answer is incomplete, but the final Decision Agent collapses many non-replies into Ambivalent rather than reserving Clear Non-Reply for explicit refusal, ignorance, or clarification.

Failure overlap is also limited. Only about 46% of the two systems' errors overlap, so the agentic decomposition is not merely duplicating the direct prompt's mistakes. It creates a different error surface: more interpretable and more sensitive to partialness, but less reliable as a classifier.

8. Conclusion

The completed system provides a reproducible D3-Agentic Prompting framework and records all intermediate agent outputs needed for interpretation. As a classifier, how-

ever, this HW4 run is substantially weaker than the earlier approaches. HW1's TF-IDF logistic regression reached about 0.45 macro F1, HW2's BERT-base fine-tuning reached 0.5561 macro F1, and HW3's best Qwen 4B few-shot prompting run reached 0.4667 macro F1 on the balanced-validation diagnostic setup. The HW4 single-agent comparator reaches only 0.3062 macro F1, and the required D3 system reaches 0.2213.

The main conclusion is therefore negative but useful: linguistic decomposition alone does not solve response-clarity classification for a very small instruction model. D3 gives a clearer audit trail and increases Ambivalent usage, but it does not make the final decision rule robust. The middle label remains the central obstacle. The direct prompt effectively refuses to detect Ambivalent, while D3 overcorrects toward Ambivalent and loses Clear Non-Reply. The final submission uses the macro-F1 selected system, but the report treats the D3 run as the core methodological result required by the assignment.

References

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